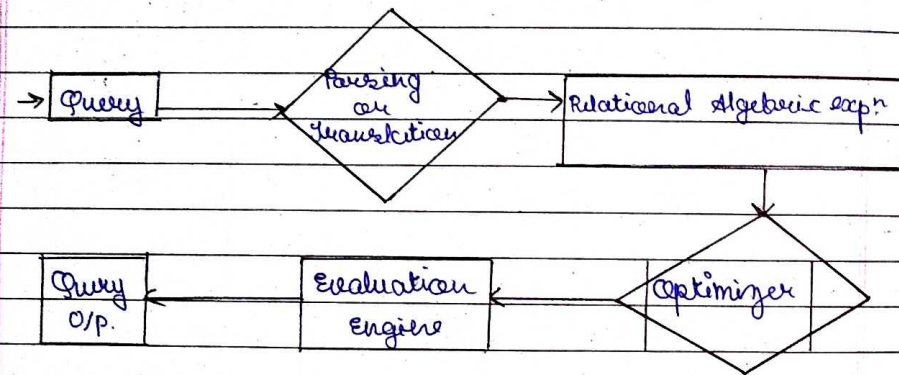


* UNIT-4 → QUERY PROCESSING *

→ Query processing is the activity performed in extracting the data from the database. In query processing it takes the various steps for fetching the data from the database.

- The steps involved:
- 1) Parsing and translation
 - 2) Optimisation
 - 3) Evaluation.



• Parsing & Translation →

- As query processing includes a certain activities for data retrieval initially the given user queries gets translated in high level DB language such as SQL.
- It gets translated into expressions that can be further used at the physical level of the file system.

• Optimisation →

- The cost of query evaluation can vary for different types of query. Although the system is responsible

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for constructing the evaluation plan, the user does not need to be sig² write their query efficiently.

- Usually a database system generate an efficient query evaluation plan which minimises its cost. This type of task performed by the database system and is known as query optimisation.

• Evaluation →

- For this with addition to the relational algebra translation it is required to translate relational algebraic expⁿ with the instructions used for specifying and evaluating each operation. Thus after translation the user query the system executes the query evaluation plan.

- Query evaluation plan → In order to fully evaluate a query the system needs to construct a query evaluation plan. The query evaluation plan may refer to the algorithm to be used for a particular index or a specific operations. Thus a query evaluation plan defines the sequence of primitive operations used for evaluating a query.
- Also referred as query execution plan.

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